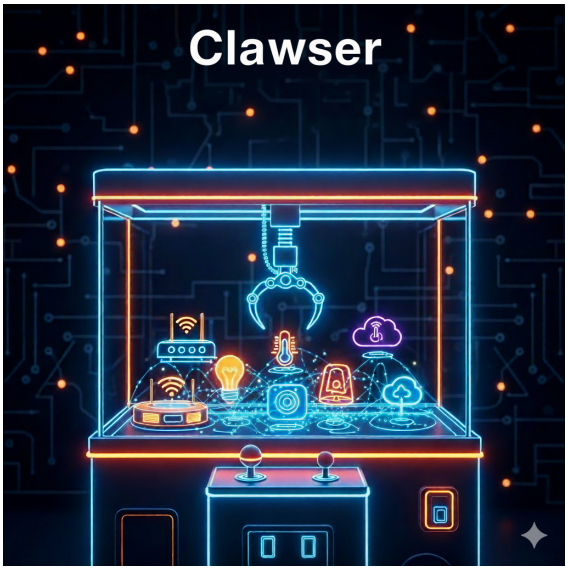
Department of Computer Science & Engineering
The University of Texas at Arlington

Project Charter
CSE 4316: Senior Design I
Fall 2025



Clawser
IOT Claw Game

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1 Problem Statement

Our project aims to construct a claw machine and the infrastructure necessary to control the machine wirelessly via a remote web application. The team will gain multiple skills applicable to future work in the industry, ranging from teamwork, communication, programming, microcontroller setup, and physical construction of the machine itself. Owners of such a project would be able to expose their claw machines to a much wider audience, while end users of this project will gain the luxury of playing a claw machine without ever needing to be physically near one.

2 Methodology

To achieve our goals, the team will need to create both physical and digital assets. The team will be segmented into three subteams to develop the hardware, backend, and frontend of the project. The hardware team will be tasked with creating the physical claw machine, which includes the claw, motors, cabinet, cameras, and sensors needed for both the physical functioning of the claw machine and connection to the online web application. The backend team will need to handle live video connections between the cameras on the computer to the application, as well as ensuring stable and responsive control of the machine to the application. Finally, the frontend team will handle creating the web application itself; they will design and implement the application where users can log in, view live video feeds of the claw machine, and most importantly, control the movements of the claw through the application.

The teams will need to complete each of their assigned duties as well as communicate with the larger team to integrate the project into a cohesive unit. To do this, the team will follow Agile principles, with work divided into sprints consisting of a sprint goal and a backlog of tasks to be accomplished during that sprint. The team will hold regular meetings to ensure the project stays on target, tasks are completed in a timely manner, and any issues that arise can be dealt with as a team.

3 Value Proposition

Our project presents a unique opportunity to showcase the integration of real time internet connectivity with a well-known real-world classic. By developing a remote-controlled claw machine that incorporates live video streaming and IoT connectivity, we are not only creating a fun and interactive system but also tapping into the growing demand for digital experiences that bridge the gap between the real and online worlds.

For industrial sponsors, this project provides an opportunity to see how their technologies, components, or platforms can be applied in creative, student-led designs. Whether it is the microcontrollers, networking solutions, video streaming modules, or the mechanical system behind the claw, this project acts as a testbed to highlight product performance in a live and highly interactive environment that has a wide range of appeal from people of all ages and backgrounds.

For university instructors and the university itself, the claw machine project represents a multidisciplinary, hands-on educational experience that brings together software, hardware, networking, and UI design. It showcases the university's commitment to fostering project-based learning and creative problem-solving skills, which are highly valued in today's technology-driven workforce. The success of this project would only strengthen UT Arlington's reputation for providing an environment where capable students can translate theory into practical and engaging solutions.

Due to the nature of the project, it has significant outreach and engagement potential. A remote claw machine game is immediately recognizable, fun, and relatable, which means that it can attract attention from a wide audience at events. The excitement of controlling a real-world machine over the internet and potentially winning a prize creates a lasting impression, making this project a memorable experience for stakeholders and spectators alike.

4 Development Milestones

- Project Charter Initial Draft - September 28th, 2025
- System Requirements Specification - October 13th, 2025
- Architectural Design Specification - November 3rd, 2025
- Demonstration of frontend interface - Q4 2025
- Demonstration of backend system - Q4 2025
- Detailed Design Specification - February 9th, 2026
- Demonstration of claw cabinet - Q1 2026
- CoE Innovation Day Poster Presentation - April 2026
- Project Charter Final Version - April 29th, 2026
- Final Project Demonstration - April 2026

5 Background

Traditional arcade claw machines have remained largely unchanged for decades, relying on mechanical components and simple analog circuitry. While these games are a nostalgic fixture in entertainment venues, they offer limited interactivity and lack integration with modern digital systems. As a result, they fail to reflect the growing relevance of Internet of Things (IoT) technologies in contemporary engineering and computer science education.

One of the primary limitations of traditional claw machines is their static, standalone nature. Without connectivity or programmability, these machines provide little educational value beyond basic mechanical operation. They do not allow for remote interaction, user data collection, or real-time system feedback, all of which are essential components in today's interconnected device landscape. This restricts their utility in teaching key concepts such as embedded systems design, wireless communication, and human-computer interaction.

In an academic setting, hands-on projects are crucial for reinforcing theoretical knowledge. However, many embedded systems projects are either too abstract or lack real-world appeal. The IoT Claw Machine Game addresses this challenge by reimagining a familiar arcade device as an engaging, educational platform. By integrating microcontrollers, sensors, and wireless modules, the system enables remote operation, real-time monitoring, and customizable gameplay, all accessible through a desktop or mobile interface.

This modernized claw machine serves as a compact, modular learning tool for students exploring IoT systems. It supports experimentation with cloud connectivity, sensor integration, and software development. Instructors can use the system to demonstrate real-world applications of networking protocols, control algorithms, and system integration.

Beyond classroom use, the project provides a foundation for future innovation. Senior Design teams can build upon the platform to explore advanced features such as machine learning-based grip optimization, multiplayer competition modes, or parallel machine communication and interaction. Industry partners and sponsors have opportunities to contribute hardware, software tools, or mentorship, gaining visibility while supporting hands-on STEM education.

By transforming a classic arcade game into a connected, interactive educational tool, this project bridges entertainment and engineering. It not only preserves the fun and challenge of the traditional claw machine but also introduces students to the possibilities of smart systems and IoT, equipping the next generation of engineers with the skills to design the connected devices of tomorrow.

6 Related Work

Claw Craziness: [1] An IoT claw machine business. We will be referring to this website as the main source of inspiration. The website has two live feeds and allows users to play the game remotely from any location. The YouTube channel offers some information on the creation of the machines, such as how they are built and an overview of how the business works. Claw Craziness offers different types of claw machine games, however, ours will be built in a classic grab-and-drop style.

MakingStuff: [3] An enthusiastic hobbyist who built a claw machine from scratch. His videos offer information for how to build the box and set up the gantry which will help us draft the hardware side of our claw machine. As the channel uses resources available at home, we will adjust plans according to what fits within the given budget.

Hackster: [2] The article provides an overview of an enthusiast's personal claw machine project, including diagrams of the machine and a link to source code for the controls. Hackster only provides a basic setup of a generic claw machine, whereas ours will be fully developed for online streaming and play.

Instructables: [4] The website shows the progress of a prototype claw machine. It includes images to

show each step of the project and a short list of items used. A video is included, providing a detailed overview of the prototype creation. This claw machine configuration does not work for the customer as it does not have the required video streaming and remote play options.
RetroBuiltGames: [5] The YouTube channel offers a video to breakdown the design and construction of an Arduino claw machine. All of the channels' references are open-source and available through GitHub repositories, offering an itemized list of materials with the costs and a detailed explanation of construction. The code used for the controls of the claw machine are also open-source and available through GitHub. However, the cost of materials for this claw machine exceeds the project budget, and this claw machine does not meet all of the sponsor's requirements for streaming and remote play.

7 System Overview

Our system enables users to play a physical claw machine over the internet. The implementation of the system can be broken down into three major components: hardware, frontend, backend.

Hardware

A physical claw machine cabinet will be constructed. This cabinet includes the claw mechanism, motors for movement of the claw, controllers to process input, and prizes. A sensor will be used to detect when a prize has been won, and cameras are mounted to the cabinet to provide live video of the machine's interior.

Frontend

The frontend is where the user will interact with the system. The user interface will be a web-based application where users can log into their accounts, view live video of the machine, and send commands to operate the claw. This approach focuses on compatibility across devices and ease of use, allowing users to interact with the system from anywhere.

Backend

The backend is the link between the other two components. It receives the input from the user interface and sends it to the motor controllers. In addition, it ensures that responses from the hardware, such as the live video, are streamed to the frontend efficiently. Low latency for input controls and video are essential here for the user experience.

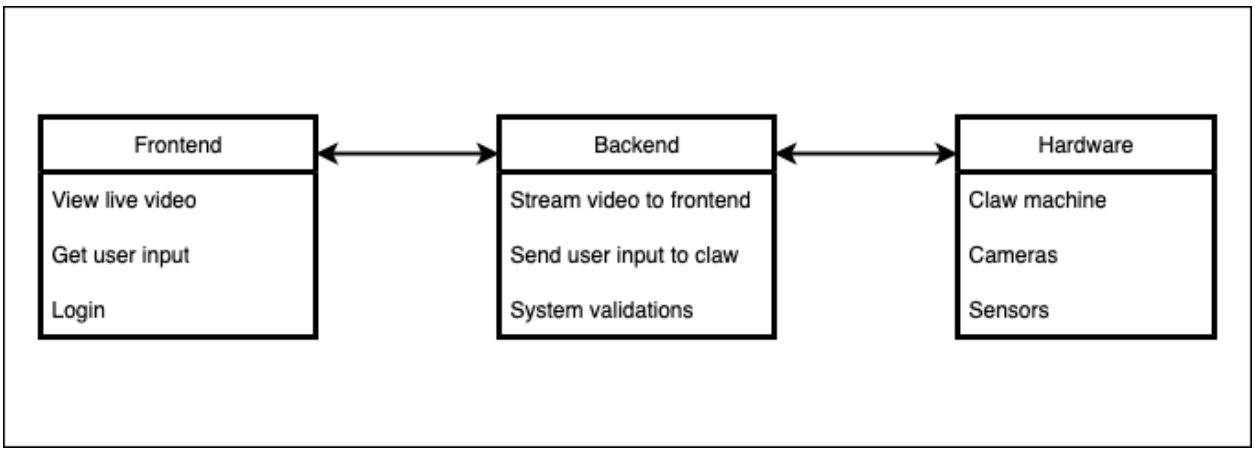

Figure 1: Diagram of Major System Components

8 Roles & Responsibilities

This project involves both hardware and software development, requiring collaboration across multiple areas.

Stakeholders

The primary stakeholders of this project are the University of Texas at Arlington's Computer Science and Engineering (CSE) Department, which is providing the funding and institutional support for development, and Professor Shawn Gieser, who serves as the project sponsor. Both parties have a vested interest in the successful completion of the project.

Points of Contact

Communication between the project sponsor and the student team is done through designated points of contact. For this project, **Talia Gacho** and **Phillip Cook** have served as the primary points of contact, ensuring that requirements, questions, and feedback flow between the sponsor and the development team. This structure provides consistency in communication throughout the project lifecycle.

Team Members and Responsibilities

The project team consists of twelve members, each assigned to one of three functional groups: backend development, frontend development, or hardware design. These groups are aligned with the project's technical requirements and ensure that work is distributed between the team. Members may be moved between groups over the course of the project as needed.

Backend Team: Carson Brown, Jesus Martinez, Talia Gacho, Jeff Hernandez
Responsibilities: Developing and maintaining the server-side logic, database management, networking protocols, and integration between the mobile app and hardware interface.

Frontend Team: Aryan Prasain, Jeff Wright, Khang Lam, Phillip Cook
Responsibilities: Designing and implementing the mobile application's user interface, ensuring usability, responsiveness, and communication with backend systems.

Hardware Team: Alexander Zacarias, David Aviles, Jordan Taylor, Josalyn Abbott
Responsibilities: Designing, constructing, and programming the claw machine hardware, including motor control, sensor integration, and connectivity with the IoT platform.

Product Ownership and Project Leadership

The role of **Product Owner** is currently held by **Talia Gacho**, who is responsible for maintaining the project vision and ensuring alignment with sponsor expectations. At this time, there are no plans to rotate this role.

9 Cost Proposal

The development of the IoT Claw Machine will require an integration of both hardware components and software services.

On the hardware side, the cabinet construction will involve materials such as lumber and plywood for the frame, as well as clear paneling (e.g., acrylic or similar materials) for the top enclosure. In addition to structural materials, the build will require various mechanical and electrical components. These include a gantry kit for the claw mechanism, along with all necessary microcontrollers, sensors, and relays.

The software portion will rely on cloud services to manage communication between the cabinet and both the frontend and backend systems. These services will also handle live video streaming and other real-time interactions.

9.1 Preliminary Budget

Component	Estimated Cost (USD)
Gantry Kit	$550.00
Cabinet Frame	$150.00
Panel(s)	$60.00
Vinyl/Decal	$80.00
Streaming Camera(s)	$120.00
Cloud Services	$250.00
Microcontroller(s)	$50.00
Sensor(s)	$25.00
Relay(s)	$20.00
Prizes	$65.00
Total Estimated Cost	$1,370.00

9.2 Current & Pending Support

Current funding for this project comes from the Computer Science and Engineering Department (CSE), which provides an initial budget of $800 per senior design grouping. As this project is comprised of two teams across two sections, this provides the complete team with an overall budget of $1600. At this time there have been no applications for additional funding; however, each member has put on reserve $50 for a total of $600 reserves should the department-provided budget not cover project expenses.

10 Facilities & Equipment

The cabinet will require a dedicated space for the storage of the cabinet. This is due to the cabinet's substantial planned physical footprint, precluding it from being easily stored in a closet. Having a dedicated space will allow for easy access to different components of the cabinet build. This includes items like the claw itself, prize stock, and control boards.

Various tools, such as hammers and screws will be necessary for cabinet modification and access to key internal components, such as board circuits and wiring. Wiring, connectors, and adaptors will also be needed for unobtrusive camera mounting. Soldering equipment is available at the Nedderman makerspace for circuit connection and repair, and the Central Library FabLab has woodworking equipment available for cabinet construction. Relevant woodworking equipment may include planer, router, and sander.

11 Assumptions

The following list contains critical assumptions related to the implementation and testing of the project.

- A suitable indoor testing location with reliable internet connectivity and adequate power outlets will be available by the 2nd sprint cycle.
- The claw machine hardware (motors, sensors, and IoT controllers) will be delivered on time and will operate according to technical specifications.
- The Wi-Fi network and cloud-based video streaming API used for remote gameplay will provide sufficient bandwidth and low latency for smooth, real-time interaction between claw game cabinet and web application.

- All team members will remain available during the project timeline and will successfully acquire the necessary technical skills within the scheduled sprints.
- Target users will have mobile devices capable of running the web application and maintaining stable internet connections for remote control of the claw machine.
- Necessary development tools, libraries, and cloud accounts will be accessible without unexpected restrictions or additional costs during the project lifecycle.

12 Constraints

The following list contains key constraints related to the implementation and testing of the project.

- Final prototype demonstration must be completed by the College of Engineering Innovation Day in mid-April 2026. The final version will be delivered to the sponsor on April 29, 2026.
- All required hardware components must be sourced from approved vendors; development tasks that rely on these parts cannot begin until delivery is completed.
- The claw machine cabinet requires a dedicated indoor workspace with reliable power and internet access.
- Total development costs must not exceed $2200. This includes the initial $800 provided per senior design group and the additional $50 per team member.
- All cabinet construction and wiring must comply with electrical and mechanical safety standards.
- Team member availability is limited by the academic calendar, course deadlines, and external personal schedules, restricting collaborative work to designated sessions.

13 Risks

The following high-level risk census contains identified project risks with the highest exposure. Mitigation strategies will be discussed in future planning sessions.

Risk description	**Probability**	**Loss (days)**	**Exposure (days)**
Problems interfacing with the gantry motherboard	0.60	8	4.8
Delays in receiving ordered parts and services	0.40	12	4.8
Software issues connecting to the cabinet and controlling it	0.50	5	2.5
Damaging the cabinet gantry/motherboard	0.25	10	2.5
Issues using cloud services for online access	0.45	3	1.35
Issues integrating webcameras into the cabinet	0.30	4	1.2
Software stack and framework issues	0.20	2	0.4
Communication errors leading to misunderstandings	0.10	2	0.2

Table 1: Overview of highest exposure project risks

14 Documentation & Reporting

14.1 Major Documentation Deliverables

14.1.1 Project Charter

The Project Charter establishes the product's high-level scope, team roles and responsibilities, usage of resources, risks, constraints, and expected deliverables. This document will be maintained and updated via the collaborative LaTeX editor Crixet. The initial version will be drafted during Sprint 1 and delivered September 28th, 2025; necessary changes may be made thereafter based on sponsor's input. The final version will be delivered April 29th, 2026, at end of project.

14.1.2 System Requirements Specification

The Systems Requirements Specification (SRS) provides a detailed breakdown of the product's functional and non-functional requirements. This document will be drafted and updated via Crixet. Revisions will be made as project scope is refined; all requirement modifications will be approved by sponsor. The initial version of the SRS will be drafted during Sprint 2 and delivered October 13th, 2025. The final version will be submitted to the sponsor on April 29th, 2026, at end of project.

14.1.3 Architectural Design Specification

The Architectural Design Specification (ADS) defines a high-level system overview of the product, outlining architectural layers, data flow, and layer subsystems. This document will be drafted and maintained in Crixet; revisions will be made upon major architectural changes. The initial draft of the ADS will be completed during Sprint 3 and delivered November 3rd, 2025. The final version will be delivered on April 29th, 2026, at end of project.

14.1.4 Detailed Design Specification

The Detailed Design Specification (DDS) provides in-depth implementation details for each product subsystem layer. This document will be collaboratively drafted and updated in Crixet. Modifications to this document will be made when detailed design elements are changed. The initial version will be drafted during Sprint 5 and delivered February 9th, 2026. The final version of the DDS will be delivered April 29th, 2026, at end of project.

14.2 Recurring Sprint Items

14.2.1 Product Backlog

Items will be added to the product backlog from the SRS based on product owner priority and estimated difficulty. Tasks given highest priority by the sponsor will be placed at the top of the product backlog, along with high-difficulty tasks and predecessor tasks that are necessary for future successor tasks to begin. Item backlog priority will be decided by team vote, taking into account stakeholder input. The product backlog will be maintained via the Planner app in Microsoft Teams, and may be exported to Excel for easy sharing with stakeholders.

14.2.2 Sprint Planning

Each two-week sprint will be planned during an in-person team meeting the week before that sprint begins. There will be four sprints in Senior Design I and four sprints in Senior Design II, for a total of eight sprints over the course of the academic year.

14.2.3 Sprint Goal

The sprint goal will be decided by the project owner, taking into account team input. The customer will provide their input on the sprint goal before each sprint formally begins, and will receive updates on

sprint goal progress halfway through each sprint.

14.2.4 Sprint Backlog

The product owner and subteam leads will decide what product backlog items make their way into the sprint backlog. Sprint tasks will be derived from the remaining highest-priority tasks in the product backlog. The backlog will be maintained via Planner scrum board and shared via Microsoft Teams.

14.2.5 Task Breakdown

Individual tasks will be assigned to team members from the sprint backlog by the product owner and subteam leads, taking into account individual preferences and skillset. Time spent on tasks will be documented via shared Excel spreadsheet for ease of sprint burn down chart creation.

14.2.6 Sprint Burn Down Charts

The team members assigned to present a given sprint's review presentations will be responsible for generating the burn down chart for that sprint. As there are two separate sections involved in this project, each sprint, there will be one team member from each section who will generate their own section-specific burndown chart. They will able to access the total amount of effort expended by each individual team member via a shared Excel spreadsheet, hosted on Microsoft Teams. The burn down chart will adhere to the burn down chart template provided on the Senior Design GitHub repository.

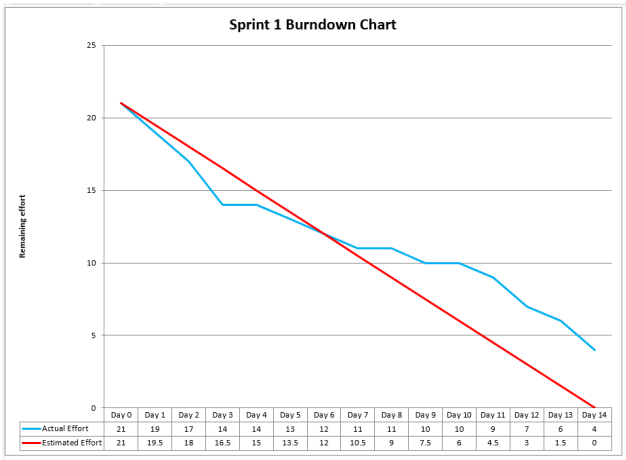

Figure 2: Example Sprint Burn Down Chart

14.2.7 Sprint Retrospective

The sprint retrospective will be addressed at the in-person team meeting the week after sprint conclusion. The team will discuss their performance and ways to improve team productivity in future sprints. Group feedback and proposed improvements will be documented in meeting minutes. Team members will document their individual retrospective in the Individual Sprint Report, due the week of sprint review.

14.2.8 Individual Status Reports

Individual members will report their status via Individual Sprint Report, submitted at the end of each sprint in PDF form to Canvas. Key items will include sprint goal, team sprint backlog, individual time expenditures, team burndown chart, individual retrospective, and peer review.

14.3 Closeout Materials

14.3.1 System Prototype

The final system prototype will include a physical claw game cabinet with functional video feed and control integration with a mobile web application. One user at a time will be able to remotely control the claw game cabinet, and will be notified of successful win upon sensor detection of prize in drop zone. The final system prototype will be demonstrated to the public during the College of Engineering Innovation Day, sometime mid-April 2026.

14.3.2 Project Poster

The project poster will include an executive summary, background, experimental setup, experimental test plan, experimental results, and conclusions. The poster will be created using the CMYK color space and printed to be 40" wide by 32" high, on glossy paper. It will be delivered in mid-April 2026, before Innovation Day.

14.3.3 Web Page

The project web page will include team and sponsor information, an abstract, background, high-level requirements, design constraints, and relevant engineering standards. It will also provide a system overview, a description of the architectural layers, a demo video, plans for future work, and links to project deliverables. It will be provided at closeout and be accessible to the public via the CSE Senior Design Student Project Repository blog.

14.3.4 Demo Video

The demo video will include footage of a user accessing the arcade cabinet control website, viewing the cabinet's camera streams, providing claw input, and acquiring a prize. The demo video will also show the physical cabinet responding to the remote input at the same time. The video will be 4-5 minutes long and will demonstrate the successful live integration of the product's website with the arcade cabinet functionality.

14.3.5 Source Code

The source code of the project will be maintained via shared team GitHub. The source code will be provided to the customer in whole, to be open-sourced to the general public at their discretion. The code will be licensed under the MIT license, with terms listed in a single README.md.

14.3.6 Source Code Documentation

The project will adhere to the documentation standards most commonly used with each constituent programming language. Documentation generator tool Doxygen will be used to generate documentation across programming languages.The final documentation will be provided in PDF form.

14.3.7 Hardware Schematics

Electronics CAD software will be used to create wiring diagrams for signal connections to controllers, motor drivers, cameras, and other peripherals as applicable.

14.3.8 CAD files

The project will involve the fabrication and modification of physical components. Mockups and schematics will be created in CAD software. Final files will be provided in OBJ form at closeout to provide details on shape, texture, color, and material information.

14.3.9 Installation Scripts

The customer will deploy software to new installations via the usage of installation scripts for the GUI front end, back end server, and for the computer directly interfacing with the arcade cabinet hardware.

14.3.10 User Manual

The customer will be provided with a digital user manual at closeout, containing diagrams, real-life pictures, and detailed explanations on product maintenance and software installation.

References

[1] ClawCraziness. We're making a 24/7 online claw machine app!, 2019.

[2] Cameron Coward. Build your own claw machine that doesn't cheat, 2025.

[3] MakingStuff. Making a claw machine, 2015.

[4] MMM. Homemade claw machine, 2016.

[5] RetroBuiltGames. Open source full size arduino claw machine for ramen noodles.